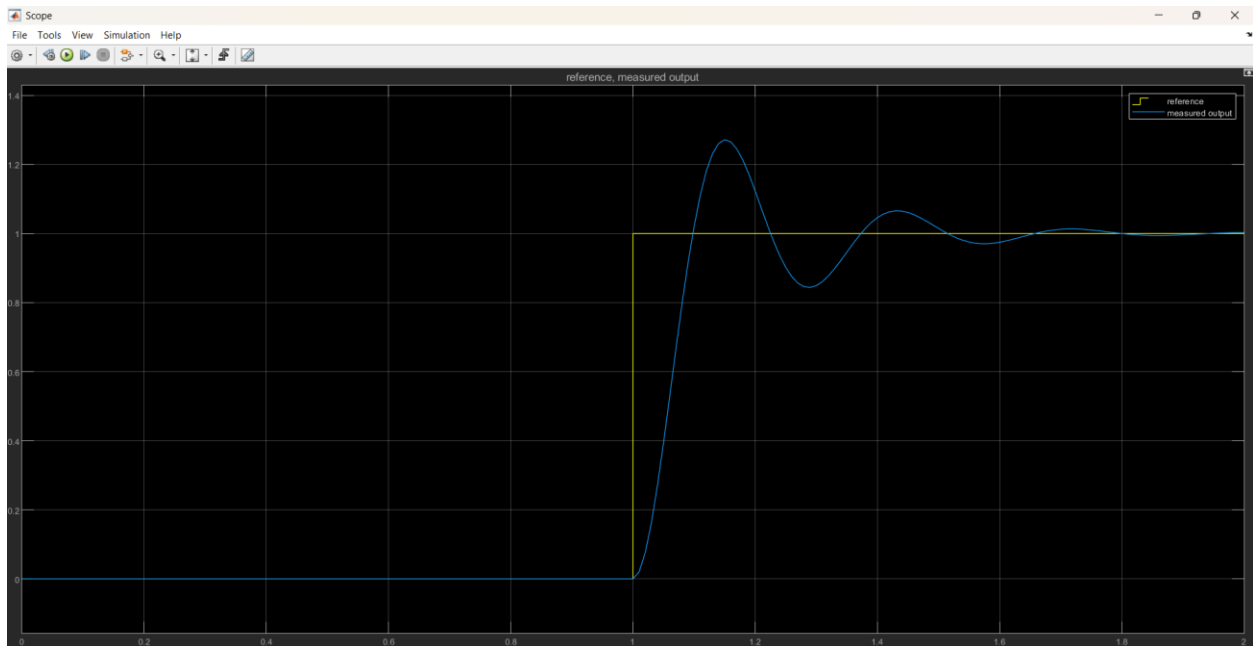
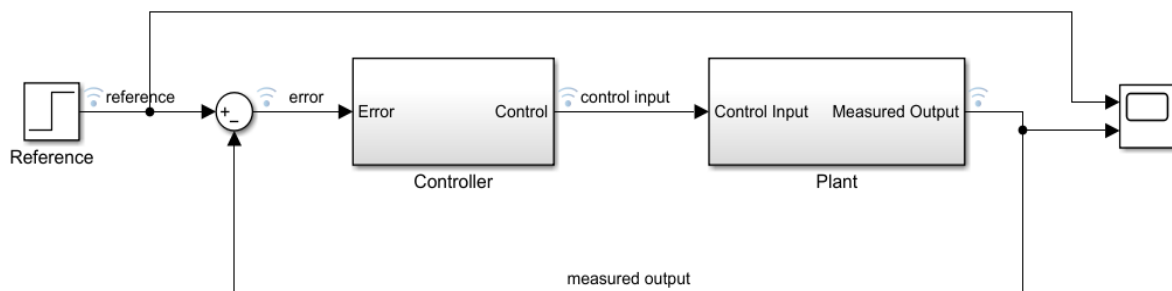


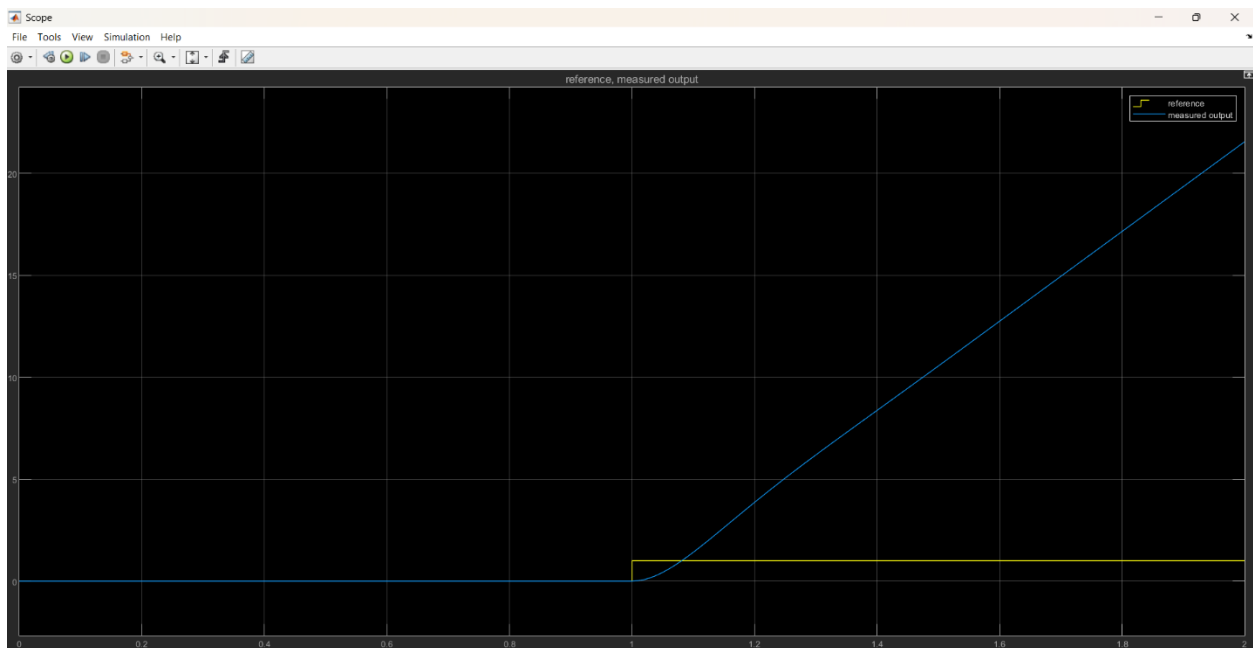
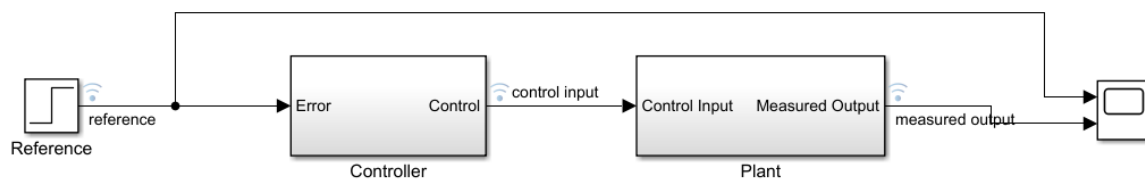
Closed Loop System

In a closed loop system, the output is continuously compared with the setpoint. Since the system receives information about the actual output via feedback, it actively works to minimize the error and make the output as close as possible to the desired value.



Open Loop System

After removing the feedback, the system becomes open loop. Since the system no longer receives information about the output, it cannot compare the result with the setpoint. Therefore, as seen from the scope, the system does not attempt to bring the output closer to reference, meaning it cannot account for errors and disturbances.



Adding Noise and Disturbance

We can test the system's noise attenuation and disturbance rejection by adding random noise and disturbances. As seen from the graph, even when spikes occur at certain intervals, the system quickly stabilizes itself. This demonstrates a high quality of disturbance rejection and effective noise attenuation provided by the feedback controller.

